Heart Disease Analysis Using Tableau

Final Project Report

1. INTRODUCTION

significant challenge to global healthcare systems. According to various health studies, cardiovascular diseases account for a large percentage of deaths each year. Rapid urbanization, sedentary lifestyles, unhealthy eating habits, excessive stress, smoking, alcohol consumption, and lack of physical activity have significantly increased the risk of heart-related illnesses. As modern lifestyles become more demanding, individuals often neglect essential health practices, which contributes to the growing prevalence of cardiovascular diseases across different age groups.

In recent years, healthcare organizations and medical institutions have started generating massive volumes of health-related data through electronic health records, medical surveys, hospital databases, and diagnostic systems. However, collecting data alone is not sufficient to understand disease patterns. Extracting meaningful insights from large datasets requires effective data analysis techniques, structured databases, and powerful visualization tools. This is where business intelligence and data analytics play an important role in transforming raw healthcare data into valuable knowledge that can assist in better decision-making.

This project focuses on analyzing heart disease data using data analytics and visualization techniques with Tableau dashboards. The primary goal of this project is to convert complex healthcare data into understandable visual insights that help identify patterns, correlations, and potential risk factors associated with heart disease. By presenting the data in the form of charts, graphs, and interactive dashboards, users can easily interpret the information and gain a clearer understanding of how different lifestyle and health factors influence cardiovascular health.

The dataset used in this project contains 4501 records and 18 attributes, representing various medical and lifestyle indicators that may contribute to heart disease. These attributes include important health-related variables such as Body Mass Index (BMI), smoking habits, alcohol consumption, sleep duration, physical activity level, diabetes status, gender, and age category. Each of these attributes plays an important role in understanding how different demographic and lifestyle factors affect the likelihood of developing heart disease.

2. DATASET DESCRIPTION

The dataset used in this project contains information about individuals and various factors that may influence the likelihood of developing heart disease. Each row represents an individual and each column represents a specific health or lifestyle attribute. The dataset contains 4501 rows and 18 columns. The target variable in the dataset is HeartDisease, which indicates whether the individual has been diagnosed with heart disease. The attributes included in the dataset represent medical conditions, lifestyle habits, demographic information, and overall health indicators.

Column Name	Description
HeartDisease	Target variable indicating presence of heart disease
BMI	Body Mass Index representing weight relative to height
Smoking	Indicates whether the individual smokes
AlcoholDrinking	Represents alcohol consumption habits
Stroke	Indicates history of stroke
PhysicalHealth	Number of days with poor physical health in a month
MentalHealth	Number of days with poor mental health in a month
DiffWalking	Difficulty walking or climbing stairs
Sex	Gender of the individual
AgeCategory	Age group classification
Race	Demographic racial category
Diabetic	Indicates diabetes condition
PhysicalActivity	Participation in physical exercise
GenHealth	General health status
SleepTime	Average hours of sleep per day
Asthma	Respiratory health condition
KidneyDisease	Indicates kidney disease
SkinCancer	Indicates skin cancer

3. IDEATION PHASE

The **Ideation Phase** of the project focuses on understanding the core problem, identifying the objectives of the study, and exploring possible solutions that can effectively address the issue. In any data analytics project, the ideation stage plays a critical role because it lays the foundation for the entire system design and analysis process. During this phase, the problem is analyzed from different perspectives, potential methodologies are discussed, and the most appropriate approach is selected based on feasibility and expected outcomes.

Heart disease is a complex health condition influenced by a wide range of factors including medical history, lifestyle habits, environmental conditions, and demographic characteristics. Factors such as unhealthy diet, smoking, alcohol consumption, lack of physical activity, obesity, and chronic diseases like diabetes significantly increase the risk of cardiovascular problems. In addition to these lifestyle factors, demographic attributes such as age group, gender, and socioeconomic conditions also contribute to the likelihood of developing heart disease. Because of the involvement of multiple interconnected variables, understanding the relationships between these factors becomes challenging when dealing with large volumes of healthcare data.

Healthcare datasets often contain thousands of records and multiple attributes representing different health indicators. When these datasets are presented in the form of raw tables or traditional spreadsheets, it becomes difficult for users to interpret patterns, correlations, and trends effectively. Traditional statistical tables and textual reports often fail to communicate insights in a clear and intuitive manner. As a result, important information may remain hidden within the dataset, making it difficult for decision-makers to identify risk factors or develop preventive strategies.

To overcome these challenges, modern data visualization and business intelligence tools are widely used to transform raw data into meaningful visual insights. Visualization platforms such as Tableau allow analysts to represent complex datasets through interactive charts, graphs, heat maps, and dashboards. These visual elements help users quickly understand patterns, identify anomalies, and explore relationships between different variables. Visual analytics also improves decision-making by presenting information in a format that is easy to interpret and interact

REQIURMENT ANALYSIS

Functional Requirements
Functional requirements describe the specific operations and tasks that the system must perform in order to achieve the project objectives.

• **Import dataset into the database**
The system should allow the heart disease dataset to be imported into a database management system. In this project, the dataset containing health indicators is stored in a MySQL database so that it can be organized and managed efficiently.

• **Perform SQL queries for data processing**
SQL queries are required to process and structure the data before visualization. This includes selecting relevant attributes, filtering records, and preparing the dataset for analysis.

• **Connect MySQL database to Tableau**
The system should establish a connection between the MySQL database and Tableau so that the data can be accessed directly for visualization and dashboard creation.

• **Create multiple visualizations**
The project should generate different types of visual representations such as bar charts, pie charts, line charts, and heatmaps. These visualizations help highlight patterns and relationships within the dataset.

• **Develop an interactive dashboard**
An interactive Tableau dashboard should be created to combine all visualizations in a single interface. The dashboard should allow users to explore the data through filters and interactive elements.

Non-Functional Requirements
Non-functional requirements describe the quality attributes and performance expectations of the system.

• **Responsive dashboard design**
The dashboard should have a clear and organized layout so that users can easily navigate and understand the information being presented.

• **Efficient data rendering**
The system should be able to display visualizations quickly and handle the dataset without performance issues.

• **Clear and understandable visualizations**
All charts and graphs should be easy to interpret, ensuring that users can quickly understand the trends and insights related to heart disease risk factors.

• **Accessibility across multiple devices**
The dashboard should be accessible on different devices such as laptops, desktops, and tablets so that users can view the analysis conveniently.

PROJECT DESIGN

The system architecture of this project explains how different tools and components work together to convert raw healthcare data into meaningful visual insights. The architecture is designed in a simple and structured way so that data can move smoothly from storage to analysis and finally to visualization. Each component in the system performs a specific role, and together they help transform the dataset into an interactive dashboard that users can explore easily.
The overall flow of the system can be represented as:

Dataset → MySQL Database → SQL Processing → Tableau Visualization → Dashboard → Tableau Public

The process begins with the heart disease dataset, which contains multiple records representing different health indicators and lifestyle factors. This dataset includes attributes such as age category, body mass index (BMI), smoking habits, alcohol consumption, sleep duration, physical activity level, and diabetes status. These variables are important because they help identify possible factors associated with heart disease.

In the first step, the dataset is imported and stored in the **MySQL database**. MySQL acts as the central storage system where the data is organized into tables. Using a database system provides several advantages, such as structured storage, easier data management, and the ability to perform queries for analysis. Instead of working directly with raw spreadsheet files, storing the dataset in a database allows the data to be accessed efficiently.

After storing the dataset in the database, the next stage involves **SQL processing**. SQL queries are used to organize and prepare the data for analysis. During this step, the data may be filtered, sorted, or grouped depending on the analysis requirements. SQL also helps in selecting relevant attributes from the dataset and removing unnecessary information. This preparation stage ensures that the data used for visualization is clean, structured, and suitable for analysis.

Once the data is prepared, **Tableau** is used as the visualization tool. Tableau connects directly to the MySQL database and retrieves the processed data. This connection allows Tableau to access the data dynamically and update the visualizations whenever changes occur in the dataset. Tableau provides various tools for creating visual representations of the data in a user-friendly way.

PROJECT PLANNING & SCHEDULING

Phase 1 – Problem Identification
The first phase of the project focused on understanding the problem and identifying the need for analysis. Heart disease is one of the major causes of death worldwide, and many factors such as lifestyle habits, medical conditions, and demographic characteristics contribute to the risk of developing cardiovascular diseases. However, analyzing these factors through raw datasets can be difficult. Therefore, the goal of this project was defined as analyzing heart disease.

Phase 2 – Data Collection
After defining the problem, the next step involved collecting a suitable dataset for analysis. The dataset used in this project contains **4501 records** and multiple attributes related to health indicators and lifestyle habits. These attributes include variables such as age category, BMI, smoking habits, alcohol consumption, sleep duration, diabetes status, and physical activity levels.

Phase 3 – Database Setup
In this phase, the dataset was imported into a **MySQL database**. Using a database management system helps in organizing and storing the data in a structured format. MySQL allows users to manage large datasets efficiently and perform various operations such as querying, filtering, and sorting data.

Phase 4 – Data Preparation
Once the dataset was stored in the database, the next step involved reviewing and preparing the data for analysis. During this stage, the dataset structure was carefully examined to ensure that all attributes were correctly defined and that the data types were appropriate. Data preparation also included verifying that there were no missing values or inconsistencies that could affect the analysis. This step ensured that the dataset was ready for visualization and interpretation.

Phase 5 – Visualization Development
After the data preparation stage, the project moved to visualization development. In this phase, **Tableau** was used to create different types of charts and graphs based on the dataset. Visualization techniques help transform numerical data into graphical representations that are easier to interpret. Various charts such as bar charts, scatter plots, and heatmaps were created to highlight relationships between different health indicators and heart disease risk factors.

Phase 6 – Dashboard Creation
Once the individual visualizations were created, they were combined into a **single interactive dashboard**. The dashboard serves as a central interface where users can view multiple visualizations together. The layout of the dashboard was designed in a way that makes the information easy to understand and visually organized. Interactive filters were also added to allow users to explore the dataset

The testing phase focused on evaluating the functionality and performance of the dashboard. During this stage, the visualizations were reviewed to ensure that the data displayed was accurate and consistent with the original dataset. The responsiveness of the dashboard and the effectiveness of filters were also tested. This step helped identify any errors or issues before finalizing the project.

Phase 8 – Documentation
The final phase involved preparing the **project documentation and report**. Documentation is an important part of any project because it explains the objectives, methodology, tools used, and results obtained. The report provides a detailed explanation of the entire development process, from data collection and database setup to visualization and dashboard creation..

FUNCTIONAL AND PERFORMANCE TESTING

Performance testing focuses on evaluating how efficiently the system handles data and how quickly it responds when users interact with the dashboard. In this project, performance testing mainly involved monitoring the **volume of data retrieved from the MySQL database** and ensuring that SQL queries were optimized for efficient data access.

Since the dataset contains **4501 records**, it is relatively manageable for visualization tools like Tableau. However, it was still important to ensure that the database queries retrieve only the necessary data required for each visualization. Optimized queries help reduce loading time and ensure that the dashboard performs smoothly without delays.

Another aspect of performance testing involved checking how quickly the dashboard updates when filters are applied. The response time of visualizations was observed to ensure that users can interact with the dashboard without experiencing slow performance.

Data Filters
Filters play an important role in making dashboards interactive and user-friendly. In this project, filters were added to allow users to focus on specific segments of the dataset. Functional testing included verifying that these filters correctly update the visualizations when applied.

Some common examples of filters used in the dashboard include filtering the data based on age category, gender, smoking status, and physical activity level. By applying these filters, users can analyze specific groups within the dataset and observe how different factors influence heart disease risk.

This interactive capability makes the dashboard more useful for exploring patterns and relationships within the data. created to derive new metrics such as risk categories or ratios.

RESULT

Different types of visualizations were used in the project to analyze the dataset from multiple perspectives. These include **trend charts, comparative bar graphs, scatter plots, and heat maps**. Each type of chart serves a different analytical purpose. For example, bar graphs help in comparing values across different categories, scatter plots help in identifying relationships between variables, and heat maps highlight concentrations or patterns within the data.

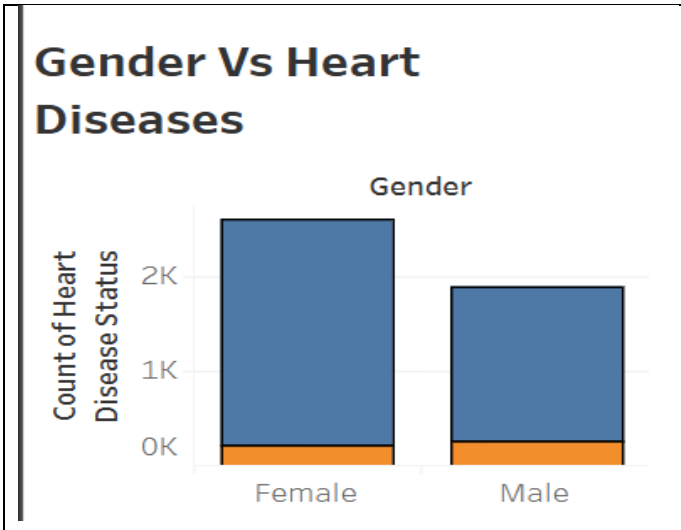
After creating individual charts, all the visualizations were combined into a **single interactive dashboard** using Tableau. The dashboard acts as the main interface where users can view multiple charts together and explore the information in an organized way. The layout of the dashboard was designed to ensure that the charts are clearly visible and easy to interpret. By presenting the data in this format, users can quickly understand important insights without needing to examine raw tables or complex numerical values.

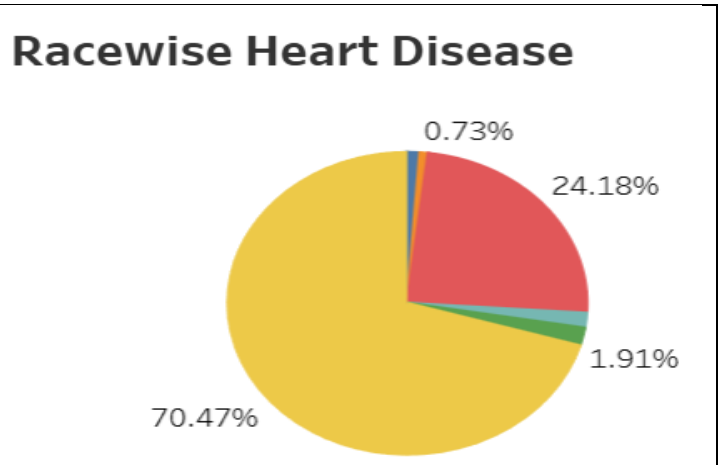
One of the key features of the dashboard is its **interactive capability**. Users can apply filters and interact with different charts to focus on specific sections of the dataset. This interactive design makes the analysis more flexible and allows users to explore the data according to their interests or requirements. For instance, users can filter the dataset by attributes such as age group, gender, smoking status, or physical activity levels to observe how these factors influence heart disease risk.

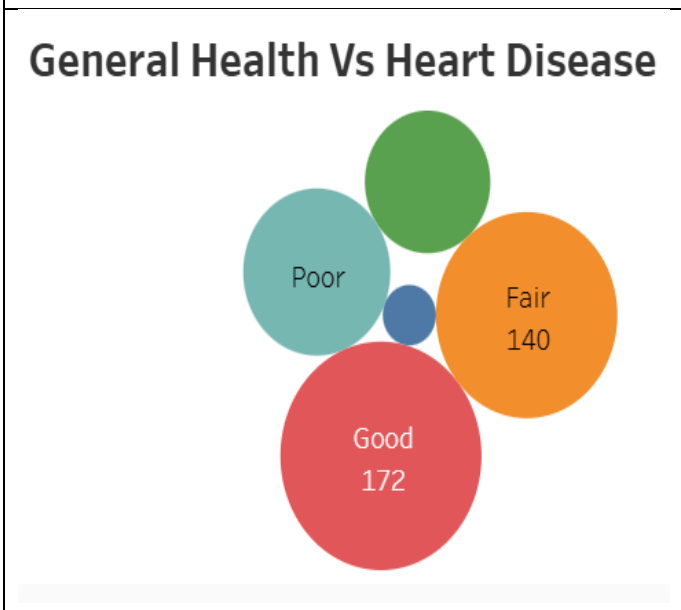
Another useful analysis presented in the dashboard is the **comparison of general health categories**. This visualization allows users to explore how individuals who report different health conditions, such as excellent, good, or poor health, are distributed within the dataset. Such comparisons help provide a broader understanding of how self-reported health status may relate to heart disease risk.

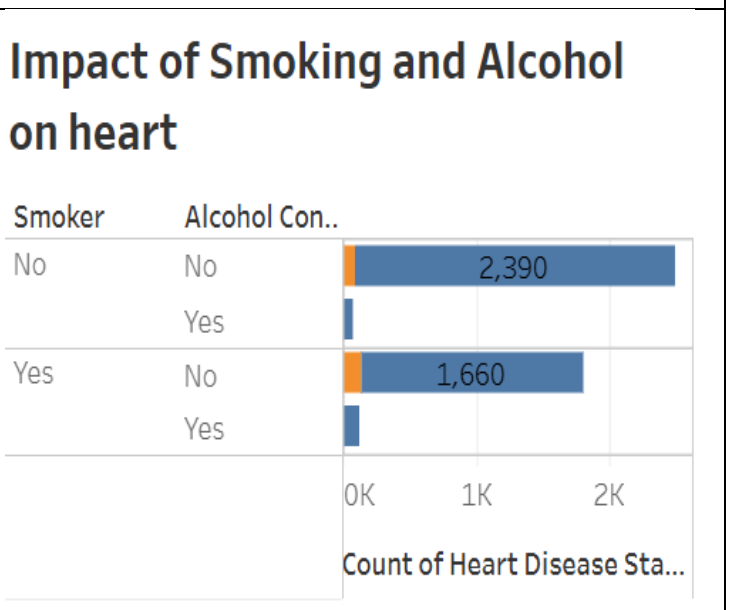
Finally, the completed dashboard was **published on Tableau Public**, which allows it to be accessed and viewed online. Publishing the dashboard makes it possible for users, instructors, and other interested individuals to interact with the visualizations through a web browser. This step also demonstrates how business intelligence tools can be used to share analytical insights with a wider audience.

OUTPUT SCREENSHOT









STORY EXPLANATION

The Tableau dashboard developed in this project includes multiple visualizations that highlight relationships between demographic factors, lifestyle habits, and the prevalence of heart disease. Each visualization focuses on a specific aspect of the dataset and together they provide a comprehensive overview of the factors associated with cardiovascular risk.

Age Category vs Heart Disease

One of the visualizations in the dashboard analyzes the relationship between age categories and the occurrence of heart disease. This chart compares different age groups to determine which groups have a higher prevalence of heart disease. The visualization shows that the likelihood of heart disease generally increases with age. Younger age groups tend to have lower reported cases, while older age categories show a higher concentration of heart disease cases. This pattern highlights age as an important factor influencing cardiovascular health.

Smoking Status and Heart Disease

One of the visualizations in the dashboard analyzes the relationship between age categories and the occurrence of heart disease. This chart compares different age groups to determine which groups have a higher prevalence of heart disease. The visualization shows that the likelihood of heart disease generally increases with age. Younger age groups tend to have lower reported cases, while older age categories show a higher concentration of heart disease cases. This pattern highlights age as an important factor influencing cardiovascular health.

Smoking Status and Heart Disease

Another visualization examines the impact of smoking habits on heart disease. Smoking is widely recognized as a significant risk factor for cardiovascular diseases. The chart compares individuals who smoke with those who do not smoke. The visualization helps identify whether smoking is associated with a higher proportion of heart disease cases within the dataset. By observing this relationship, users can better understand how lifestyle choices such as smoking may contribute to long-term health risks.

BMI and Heart Disease Relationship

The dashboard also includes a visualization that explores the relationship between Body Mass Index (BMI) and heart disease. BMI is a commonly used indicator of body weight relative to height and is often associated with health risks such as obesity and cardiovascular conditions. Through the visualization, users can observe whether individuals with higher BMI values show greater occurrence of heart disease. This type of analysis helps illustrate the importance of maintaining a healthy body weight as part of overall cardiovascular health management.

Overall Insight from the Dashboard

When all the visualizations are viewed together within the dashboard, they present a clear story about the factors that may influence heart disease. The visualizations highlight how age, lifestyle habits, body health indicators, and physical activity levels interact and contribute to cardiovascular risk patterns. Instead of analyzing raw data tables, users can easily interpret these insights through graphical representations.

WEB INTEGRATION

Web integration refers to the process of publishing the developed dashboard on the internet so that it can be accessed through a web browser. In this project, the interactive dashboard created using Tableau was integrated with a web platform so that users can view and interact with the visualizations online. This allows the dashboard to be shared easily with instructors, researchers, and other users without requiring them to install any specialized software.

The integration process begins after the dashboard is finalized in Tableau. The completed dashboard is published to Tableau Public, which is an online platform that hosts Tableau visualizations. Publishing the dashboard on this platform generates a unique web link that can be shared with others. Once the dashboard is available online, it can be accessed from any device with an internet connection.

To integrate the dashboard with a website, an embedded HTML code provided by Tableau Public can be used. This code allows the dashboard to appear directly inside a webpage so that users can interact with the charts without leaving the website. The embedded dashboard retains all interactive features such as filters, tooltips, and dynamic visual updates.

The general process of web integration includes the following steps:

The dashboard is created and finalized in Tableau.

The dashboard is published to Tableau Public.

A shareable link or embed code is generated.

The embed code is added to a webpage using HTML.

Users can access and interact with the dashboard through the website.

Through this integration, the project becomes more accessible and user-friendly. Instead of viewing static reports, users can interact with the dashboard directly through a web interface. This approach improves the usability of the project and allows the insights generated from the dataset to reach a wider audience.

Web integration also demonstrates how business intelligence tools can be combined with web technologies to create interactive data-driven applications. By making the dashboard available online, the project highlights the potential of data visualization platforms to support real-time access to analytical insights.

ADVANTAGES & DISADVANTAGES

Every data analytics project has certain strengths and limitations. Understanding these aspects is important because it helps evaluate how effective the system is and also identifies areas where improvements can be made in the future. In this project, the use of data visualization and dashboard technology offers several advantages in analyzing healthcare data, but there are also some limitations related to the scope of the dataset and the analytical techniques used.

Advantages

• Interactive exploration of healthcare data
• Easy identification of lifestyle risk factors
• Visual representation improves understanding
• Helps healthcare professionals analyze patterns
• Supports data-driven decision making

DIisadvantages

• Analysis limited to available dataset
• No predictive modeling included
• Requires internet access for Tableau Public dashboards

CONCLUSION

This project demonstrates how business intelligence tools can be effectively used to analyze healthcare datasets and present meaningful insights through visual analytics. In the modern healthcare environment, large volumes of medical data are generated every day, but this information is only useful when it can be properly analyzed and interpreted. By applying data management and visualization techniques, raw healthcare data can be transformed into valuable knowledge that supports better understanding of disease patterns and risk factors.

In this project, the integration of MySQL database management with Tableau visualization tools played an important role in organizing and analyzing the heart disease dataset. The dataset, which contains thousands of records and multiple health-related attributes, was first stored and managed using MySQL. SQL queries were then used to review and structure the data before it was used for visualization. This process ensured that the data was properly organized and ready for analysis.

The next stage involved the use of Tableau to create various charts and graphical representations of the dataset. Different visualization techniques such as bar charts, scatter plots, and heat maps were used to highlight patterns within the data. These visualizations made it easier to observe relationships between variables such as age group, body mass index (BMI), smoking habits, physical activity levels, and overall health conditions.

All of these visual elements were combined into a single interactive dashboard, which serves as the main output of the project. The dashboard allows users to interact with the data through filters and dynamic visualizations. This interactivity enables users to focus on specific categories within the dataset and analyze how different lifestyle factors influence the risk of heart disease.

The project also demonstrates the importance of data visualization in simplifying complex information. Instead of analyzing large tables of numerical values, users can quickly understand trends and patterns through graphical representations. This makes the analysis more accessible not only to data analysts but also to healthcare professionals and decision-makers who may not have a strong technical background in data science.
about preventive healthcare strategies.

FUTURE SCOPE

Although the project successfully demonstrates the use of business intelligence tools for analyzing heart disease data, there are several opportunities to further improve and expand its capabilities. Future enhancements can make the system more advanced, more accurate in analysis, and more useful for healthcare professionals and researchers. By incorporating additional technologies and larger datasets, the project can evolve into a more powerful healthcare analytics platform.

One possible improvement is the **integration of real-time healthcare datasets**. In the current project, the analysis is performed on a static dataset containing previously collected records. In the future, the system could be connected to real-time healthcare databases or hospital management systems. This would allow the dashboard to update automatically as new patient data becomes available. Real-time data analysis can help healthcare professionals monitor trends more effectively and respond quickly to emerging health risks.

Another potential enhancement is the **implementation of machine learning models for prediction**. The current system focuses mainly on descriptive analysis and visualization of existing data. However, predictive analytics could be added to estimate the likelihood of heart disease based on different health indicators. Machine learning algorithms such as decision trees, logistic regression, or classification models could analyze historical data and predict possible risk levels for individuals. This would make the system more valuable for preventive healthcare planning.

The **development of mobile-friendly dashboards** is another area that could improve the usability of the project. While the dashboard can currently be accessed through web browsers, optimizing it for mobile devices would allow users to interact with the data more conveniently. Mobile compatibility would enable healthcare professionals, researchers, and policymakers to view insights and perform analysis directly from smartphones or tablets.
size for more comprehensive analysis

APPENDIX

Dataset link :

https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

Tableau dashboard link :

https://public.tableau.com/views/HeartDiseaseAnalysis_17728812537300/DashboardFinal?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Story link :

https://public.tableau.com/views/HeartDiseaseAnalysisStory_17728816032560/HeartDiseaseStory?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GitHub link :

https://github.com/durvankmali/Heart-Disease-Analysis-Using-Tableau

Demo video link :

https://drive.google.com/drive/folders/1zgTPcEgppeKaXDXRAzXr5p0_n2yohCpX?usp=sharing